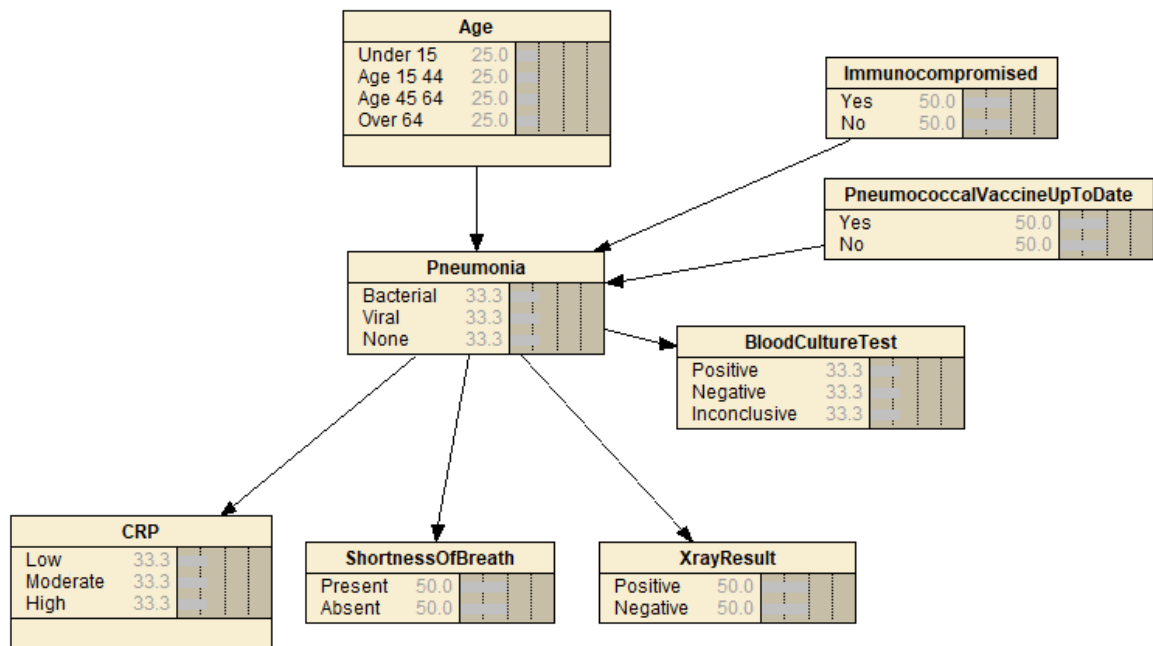


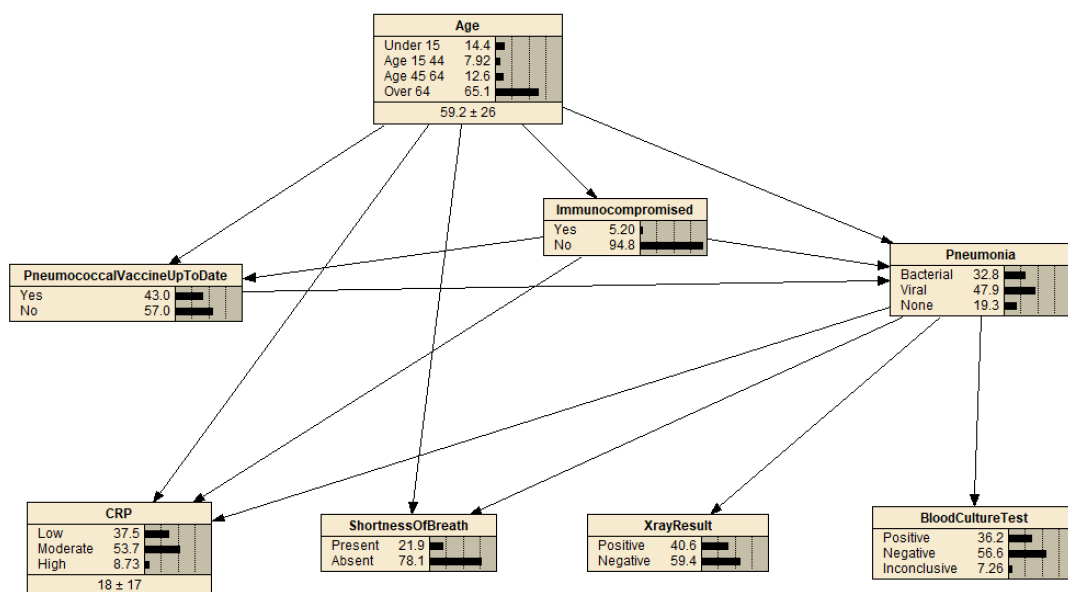
Assignment 2 – Knowledge Representation and Reasoning with Bayesian Networks

(Michael Wang, 33899835)

Question 1



Question 2

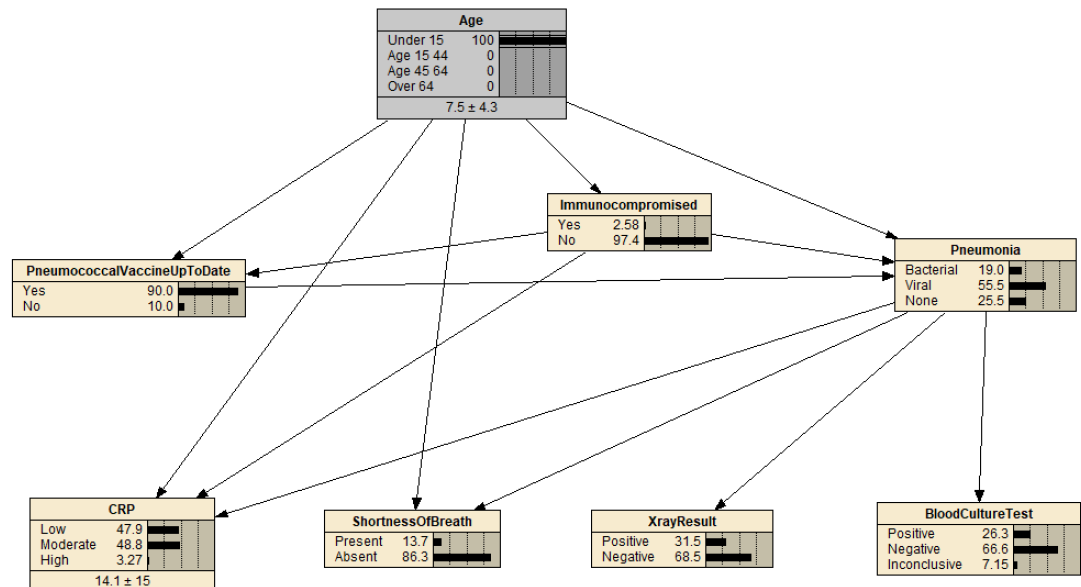


Variable Dictionary:

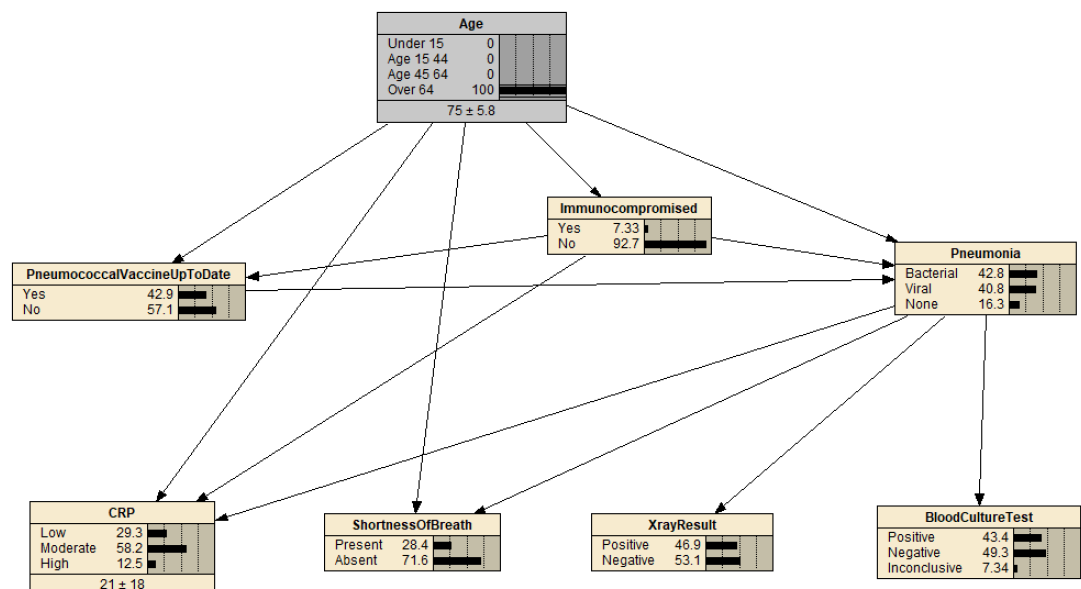
ID	Description	States	Parents	Parameterisation Source
Age	Patient's age in years.	Under 15 [0, 15), Age 15-44 [15, 45), Age 45-64 [45, 65), Over 64 [65, Infinity).	None	Data files: riskfactors, symptom, medtest
Immunocompromised	Whether the patient has a condition causing weakened immunity.	Yes, No.	Age	ME (weight 20), Data files: riskfactors, symptom
PneumococcalVaccineUpToDate	Whether the patient is up to date with pneumococcal vaccination.	Yes, No.	Age, Immunocompromised	ME (weight 20)
Pneumonia	The patient's true pneumonia status and type.	Bacterial, Viral, None	Age, Immunocompromised, PneumococcalVaccineUpToDate	ME (weight 20), Data files: riskfactors, symptom, medtest
CRP	C-reactive protein, a sign of bacterial pneumonia.	Low [0, 5), Moderate [5, 40), High [40, Infinity).	Pneumonia, Age, Immunocompromised	Data file: symptom
ShortnessOfBreath	Whether the patient presents with shortness of breath.	Present, Absent	Pneumonia, Age	Data file: symptom
XrayResult	Result of a chest X-ray, giving an indication of pneumonia.	Positive, Negative	Pneumonia	LIT (weight 60), Data file: medtest
BloodCultureTest	Result of a blood culture test, identifying causative bacteria in the blood.	Positive, Negative, Inconclusive	Pneumonia	LIT (weight 60), Data file: medtest

Question 3

- i) Given the age is under 15, the probability of the boy having bacterial pneumonia, viral pneumonia and no pneumonia are **0.19**, **0.555** and **0.255** respectively.

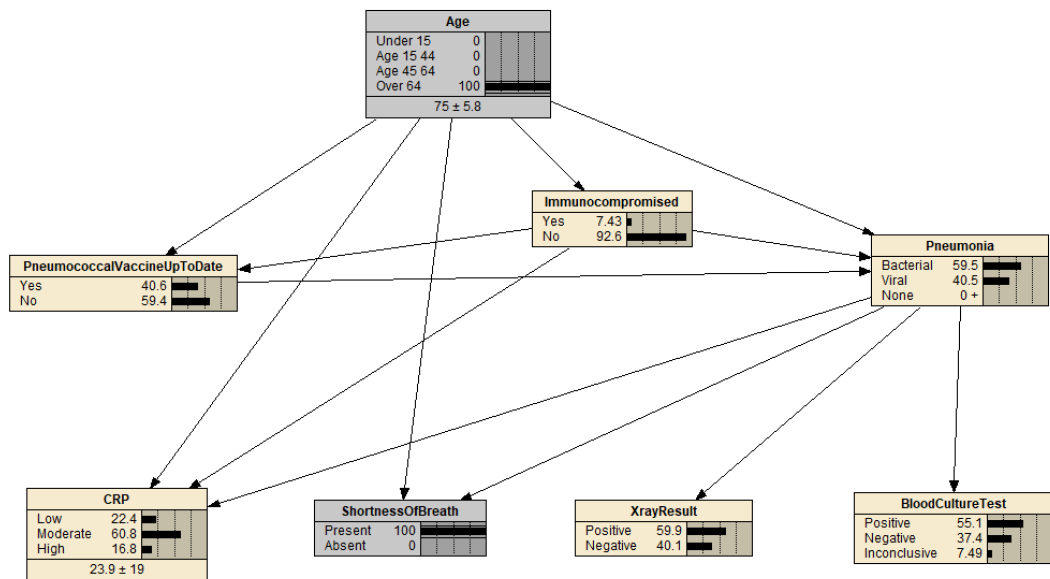


Given the age is over 64, the probability of the woman having bacterial pneumonia is **0.428**.

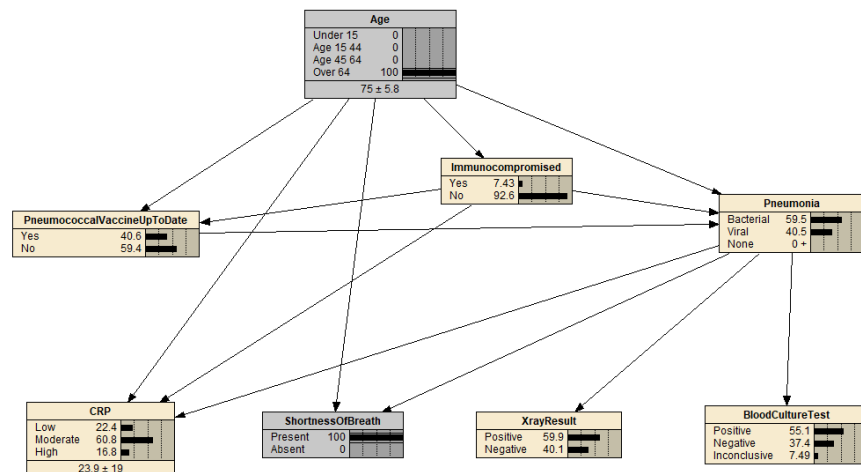
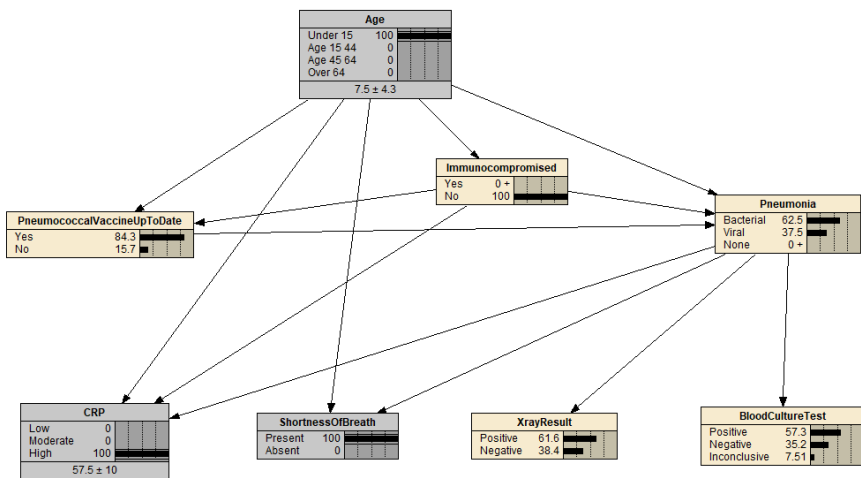


Hence, the 72-year-old woman is more likely to have bacterial pneumonia, by **0.238**. The chance of the boy having any type of pneumonia is **0.745**.

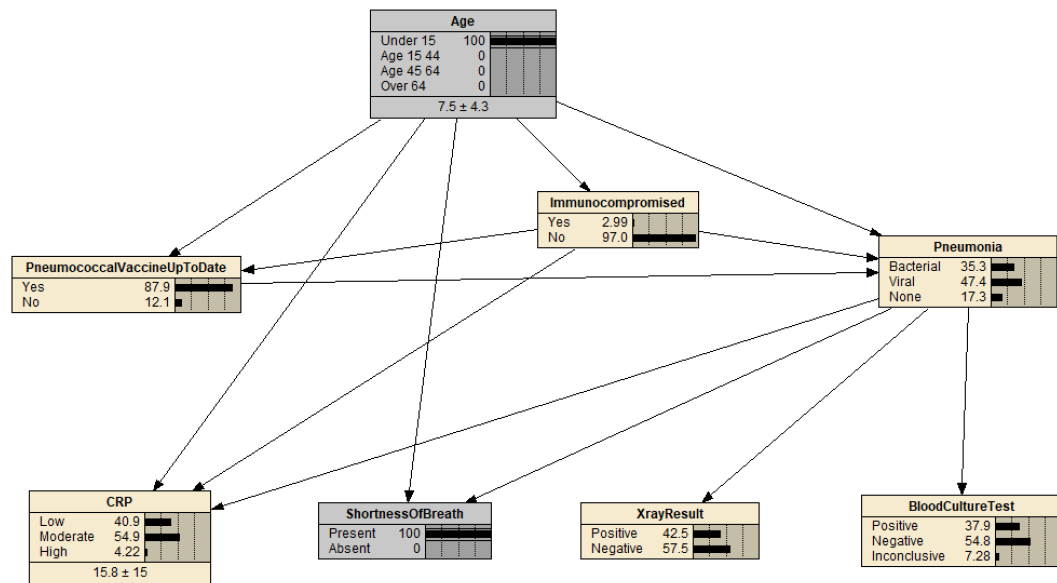
- ii) a) The woman's predicted CRP level has a mean of **23.9** and a standard deviation of **19**.



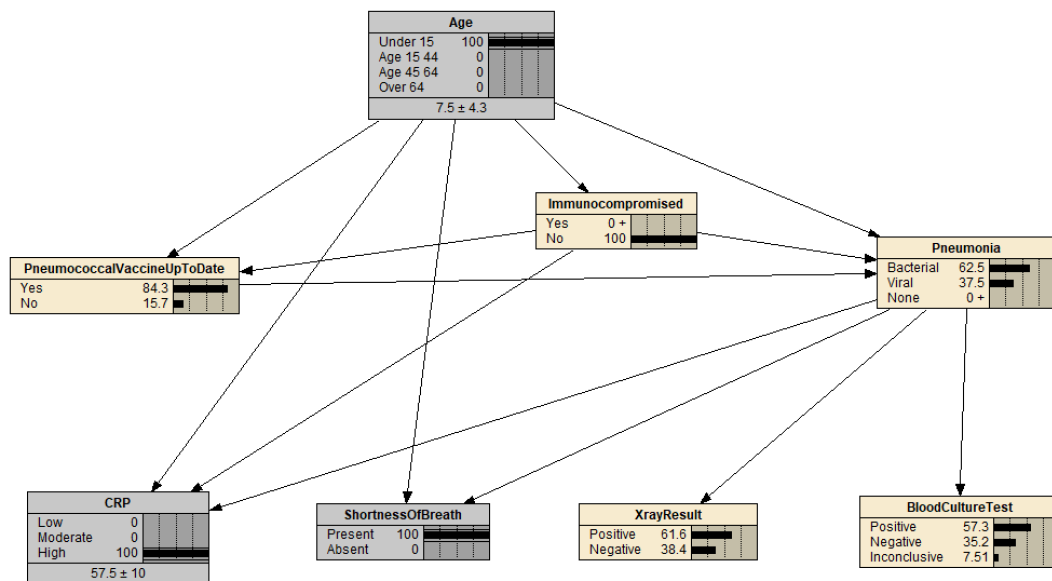
- b) Given all the information, the probability of bacterial pneumonia for the boy is **0.625** and for the woman is **0.595**.



c) Before observing CRP, the probability that the boy is immunocompromised is **0.0299**.

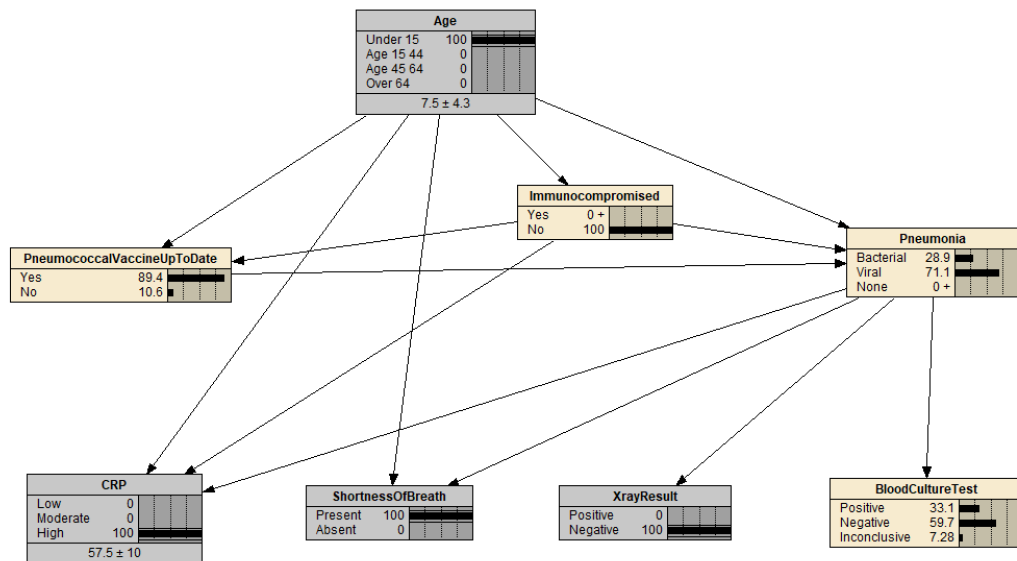


After observing CRP, the probability that the boy is immunocompromised is approximately **0 (0+)**.

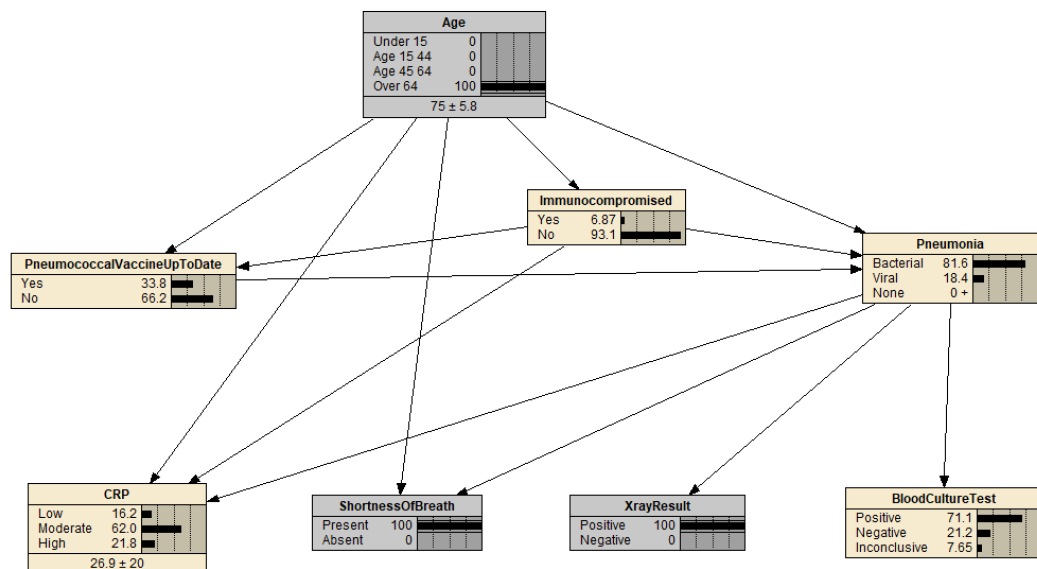


The reason for this is that in the trained CPT for CRP, the under 15/immunocompromised = Yes rows assign almost no probability to High CRP, the evidence then propagates back along the arc and decreases the probability of immunocompromised = Yes.

- iii) Given the X-ray result is negative, the probability of the boy having bacterial pneumonia and viral pneumonia is **0.289** and **0.711** respectively.



Given the X-ray result is positive, the probability of the woman having bacterial pneumonia and viral pneumonia is **0.816** and **0.184** respectively.



Hence, the woman is more likely to have bacterial pneumonia.

Question 4

- i) a) The entropy of the pneumonia node when there is no evidence is **1.494**.

Sensitivity of 'Pneumonia' to a finding at another node:

Node	Mutual Info	Percent	Variance of Beliefs

Pneumonia	1.49376	100	0.4036386
BloodCultureTest	0.41176	27.6	0.0788160
XrayResult	0.31029	20.8	0.0438091
CRP	0.17446	11.7	0.0293330
ShortnessOfBreath	0.07520	5.03	0.0068110
Age	0.07076	4.74	0.0105323
PneumococcalVaccineUpToD	0.02466	1.65	0.0052815
Immunocompromised	0.00092	0.0615	0.0000632

- b) **BloodCultureTest** would provide the most information about pneumonia on its own.

- i) The mutual information of this variable with pneumonia is 0.412 bits, which is 27.6% of the pneumonia node's entropy.
 ii) 0.412 bits, same as the expected reduction in entropy in part i).

- c) Immunocompromised provides the least information about pneumonia. Its mutual information with pneumonia is 0.00092 bits.

The reason is because about 95% of the patients are not immunocompromised, so if we are told that the patient is not immunocompromised, knowing it doesn't tell me much as this was what I expected.

On the other hand, if I was told the patient is immunocompromised, the probability of bacterial pneumonia, viral pneumonia and no pneumonia is $(130/385 = 0.338)$, $(190/385 = 0.493)$ and $(65/385 = 0.169)$ respectively, which is very close to if we have not observed it $(0.328, 0.479, 0.193)$, meaning we wouldn't have learned much anyways.

Age	Immunocompromised	PneumococcalVaccineUpToDate	Bacterial	Viral	None
Under 15	Yes	Yes	10.985	15.308	8.3037
Under 15	Yes	No	13.055	3.2367	5.3769
Under 15	No	Yes	89.925	322.36	141.46
Under 15	No	No	31.332	27.838	20.082
Age 15 44	Yes	Yes	1.5261	13.792	5.1018
Age 15 44	Yes	No	3.0783	12.39	5.1526
Age 15 44	No	Yes	2.6524	36.398	13.166
Age 15 44	No	No	43.058	191.86	75.242
Age 45 64	Yes	Yes	1.5334	13.874	5.1384
Age 45 64	Yes	No	3.0999	12.499	5.2078
Age 45 64	No	Yes	3.634	50.425	18.404
Age 45 64	No	No	48.629	326.39	117.31
Over 64	Yes	Yes	53.299	111.96	22.789
Over 64	Yes	No	44.617	6.9137	9.1882
Over 64	No	Yes	286.58	567.52	169.44
Over 64	No	No	818.91	457.65	255.32

- ii) CRP, PneumococcalVaccineUpToDate and ShortnessOfBreath now provide more information than before. Age and BloodCultureTest now provide less information than before. Some variables now have less mutual information with pneumonia because:
1. The evidence already provides enough information about these variables, knowing these would not give us much new information. For example, age would not tell us much because immunocompromised patients are concentrated in the elderly.
 2. The variable is bad at separating the remaining types of pneumonia. Knowing a negative X-ray result largely rules out the chance of a bacterial pneumonia and BloodCultureTest is bad at separating viral and no pneumonia as their rows are nearly identical.

Sensitivity of 'Pneumonia' to a finding at another node:

Node	Mutual Info	Percent	Variance of Beliefs

Pneumonia	1.26410	100	0.3110656
CRP	0.50777	40.2	0.0931164
BloodCultureTest	0.20299	16.1	0.0220889
ShortnessOfBreath	0.08640	6.83	0.0055955
PneumococcalVaccineUpToD	0.07648	6.05	0.0112352
Age	0.00871	0.689	0.0016530
XrayResult	0.00000	0	0.0000000
Immunocompromised	0.00000	0	0.0000000

- iii) a) The error rate on the test data is 29.82%
- b) The confusion matrix is shown below. It performs worst on None, only 2 of 22 healthy patients are correctly identified.
- c) The log score is 0.5571.

For Pneumonia:

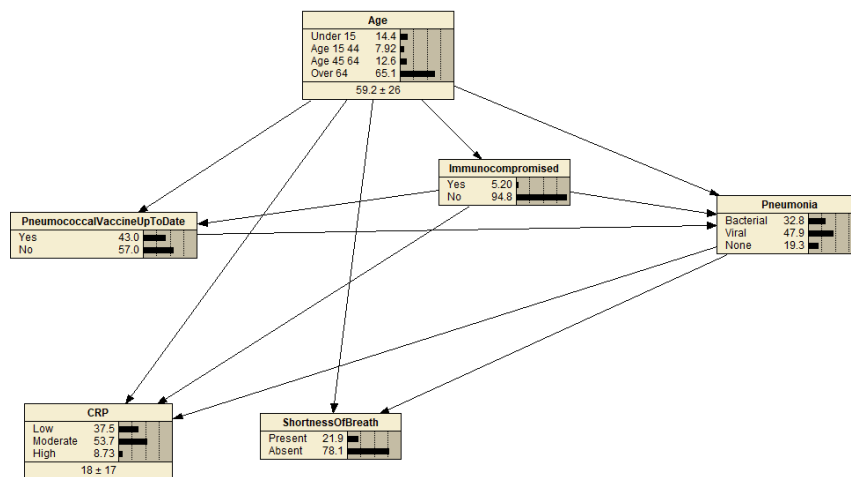
Confusion:

.....Predicted.....			Actual
Bacter	Viral	None	
-----	-----	-----	-----
35	3	1	Bacterial
7	43	3	Viral
4	16	2	None

Error rate = 29.82%

Scoring Rule Results:
 Logarithmic loss = 0.5571
 Quadratic loss = 0.3368
 Spherical payoff = 0.8011

iv) **Model 1:**



Model 2:

Pneumonia	
Bacterial	32.8
Viral	47.9
None	19.3

- a) Model 1 error rate: 39.47%
 Confusion matrix: see below.
 Log score: 0.8084

```

For Pneumonia:
-----

Confusion:
.....Predicted.....
Bacter  Viral  None  Actual
-----  -----
      28    11     0  Bacterial
      12    41     0   Viral
       3    19     0   None

Error rate = 39.47%

Scoring Rule Results:
Logarithmic loss = 0.8084
Quadratic loss   = 0.497
Spherical payoff = 0.7028
  
```

- Model 2 error rate: 53.51%
 Confusion matrix: see below.
 Log score: 1.041

```

For Pneumonia:
-----

Confusion:
.....Predicted.....
Bacter  Viral  None  Actual
-----  -----
       0    39     0  Bacterial
       0    53     0   Viral
       0    22     0   None

Error rate = 53.51%

Scoring Rule Results:
Logarithmic loss = 1.041
Quadratic loss   = 0.63
Spherical payoff = 0.6083
  
```

Compared to iii), both reduced models perform worse than the full model. The error rate rises from 29.82% to 39.47% when the two test nodes are removed, and to 53.51% for the one-node model, while logarithmic loss rises from 0.5571 to 0.8084 to 1.041. The confusion matrices show that removing the X-ray and blood culture wipes out the 'None' class entirely, with Model 1 never predicting that state at all against the full model's 2 of 22. Model 2 predicted Viral for all 114 cases and therefore misclassifying every bacterial and no-pneumonia patient.

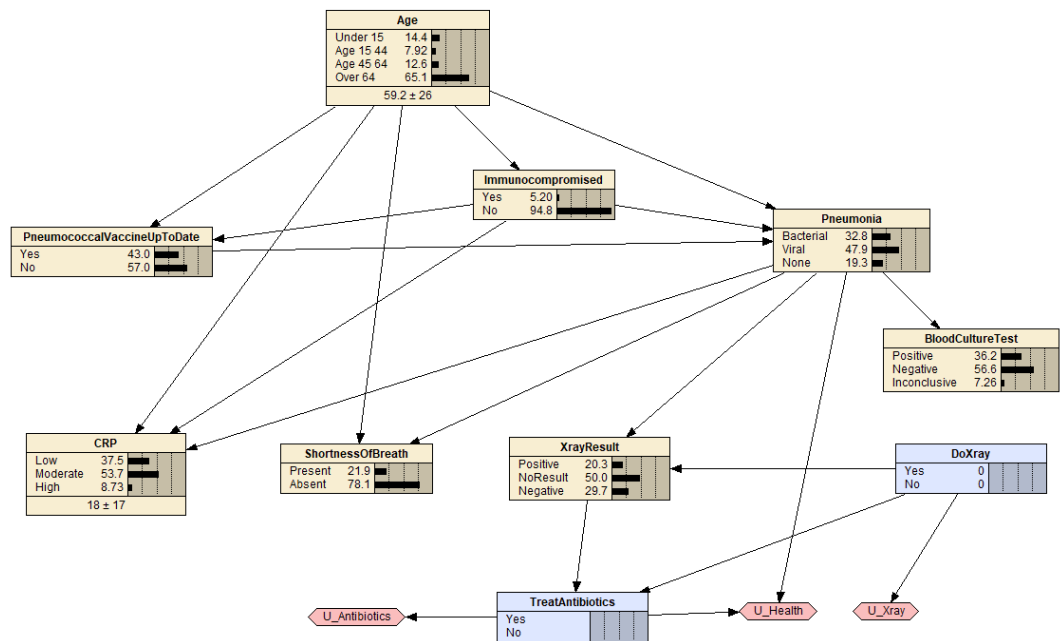
- b) The full model tells us more information than a model that simply guesses the most common answer, evident in the 29.82% error rate compared to the 53.51% error rate. The log score improves in the same way from 1.041 down to 0.5571, meaning the confidence it places on the correct diagnosis rises from roughly 35% to 57%.

Most of the improvement comes from the two test nodes. Taking away the two test nodes cost more than taking away all five of the other nodes together.

A downside of the model is that it is only good at telling bacterial pneumonia from viral pneumonia, but bad at separating a healthy patient from one with pneumonia. This is evident in the model only able to find 2 out of 22 who had no pneumonia, and without the two tests it finds none. Hence, this is useful for working out which kind of pneumonia someone has but not for deciding whether they have pneumonia.

Question 5

i) Modified Network:



ii) Let:

$b = P(\text{Bacterial} \mid \text{Negative})$ if negative result, $P(\text{Bacterial} \mid \text{Positive})$ otherwise

$v = P(\text{Viral} \mid \text{Negative})$ if negative result, $P(\text{Viral} \mid \text{Positive})$ otherwise

$n = P(\text{None} \mid \text{Negative})$ if negative result, $P(\text{None} \mid \text{Positive})$ otherwise

X = cost and harm of the X-ray

A = cost and harm of the antibiotics

B = harm of bacterial pneumonia left untreated

V = harm of viral pneumonia

U_{Health} : (Bacterial, treat) = 0, (Bacterial, no treat) = $-B$, (Viral, either) = $-V$, (None, either) = 0.

$$EU(\text{treat}) = -X - A + [b * 0 + v * -V + n * 0] = -X - A - v * V$$

$$EU(\text{no treat}) = -X + [b * -B + v * -V + n * 0]$$

$$= -X - b * B - v * V$$

$$EU(\text{treat}) - EU(\text{no treat}) = b * B - A$$

➔ Treat when $b > A/B$

After a positive X-ray result, treat when $A/B < b = 0.253$.

After a negative X-ray result, not treat when $A/B > b = 0.0151$

We want to choose A, B such that $0.0151 < A/B < 0.253$.

Choose $A = 20, B = 200$.

Choose $V = 20$ to show that untreated bacterial is worse than viral.

For X-ray utility, let P be the chance the X-ray reads positive give the patient took it.

$$EU(\text{no } x - \text{ray}) = -b_n * B - v * V$$

$$EU(x - \text{ray}) = -X + P(-A - v * V) + (1 - P)(-b * B - v * V)$$

$$EU(x - \text{ray}) - EU(\text{no } x - \text{ray}) = P(b * B - A) - X$$

Hence, we should x-ray when $X < P(b * B - A) = 0.216 * (0.253 * 200 - 20) = 6.61$

Choose $X = 3$.

Overall utility table:

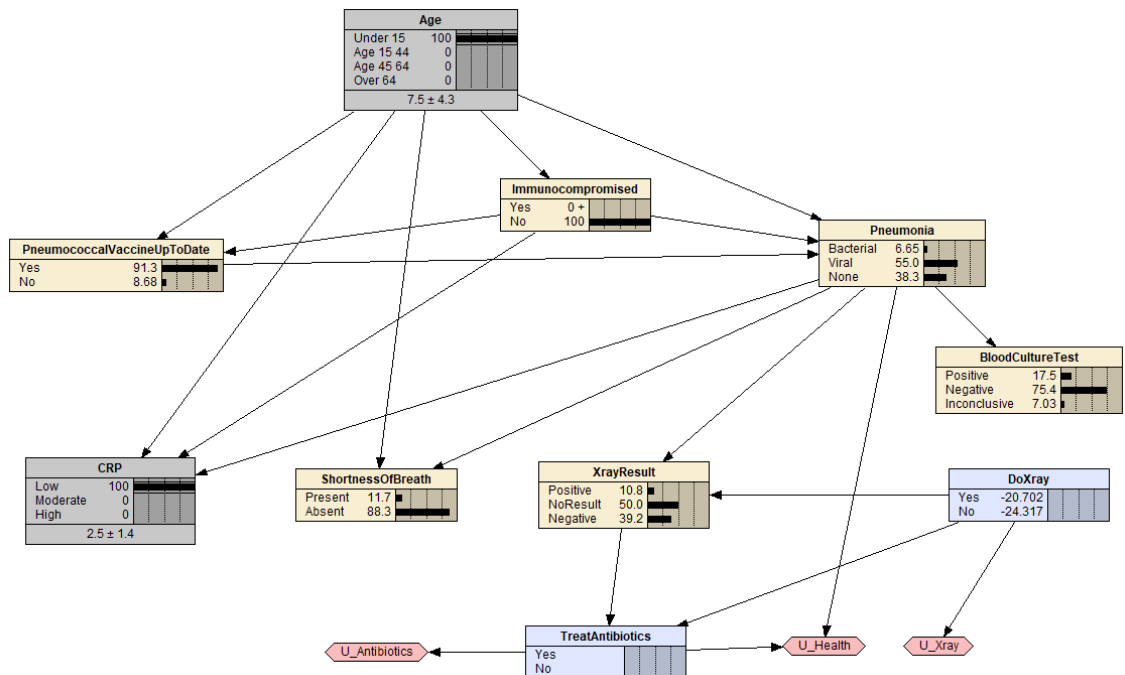
TreatAntibiotics		U_Antibiotics
Yes		-20
No		0

Pneumonia	TreatAntibiotics	U_Health
Bacterial	Yes	0
Bacterial	No	-200
Viral	Yes	-20
Viral	No	-20
None	Yes	0
None	No	0

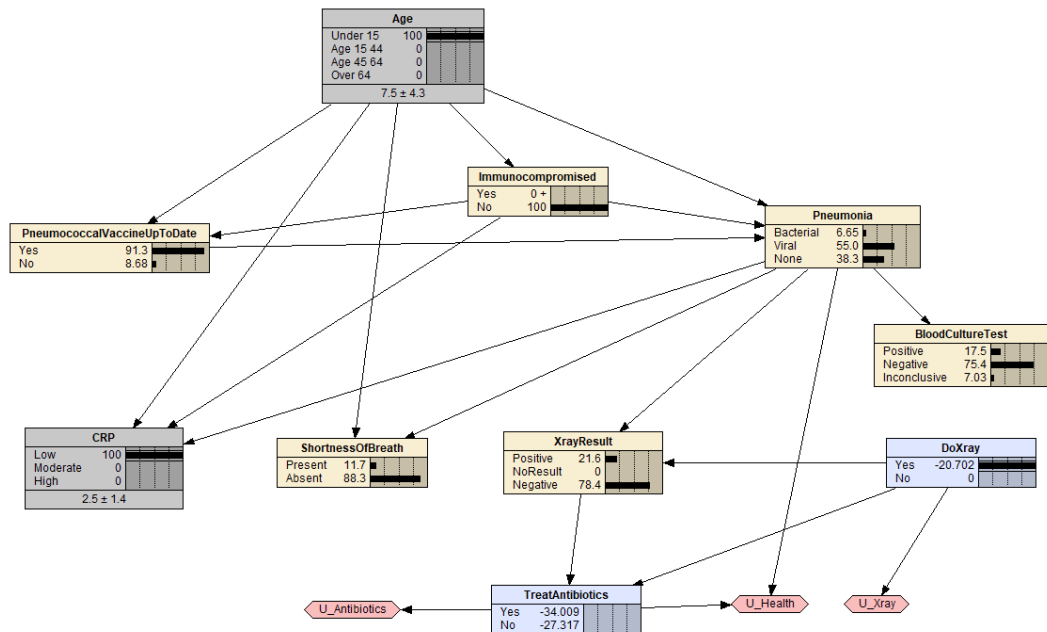
DoXray	U_Xray
Yes	-3
No	0

Question 6

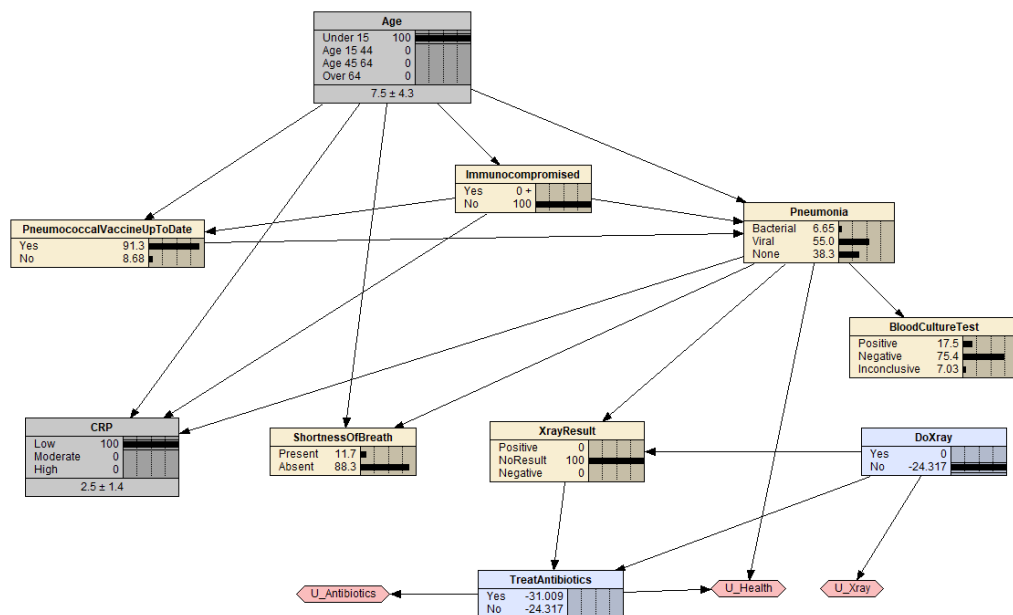
- i) The expected utility for doing the X-ray test is -20.702, and for not doing the X-ray test is -24.317.



Given the patient did X-ray, the expected utility for taking antibiotics is -34.009 and for not taking the antibiotics is -27.317.

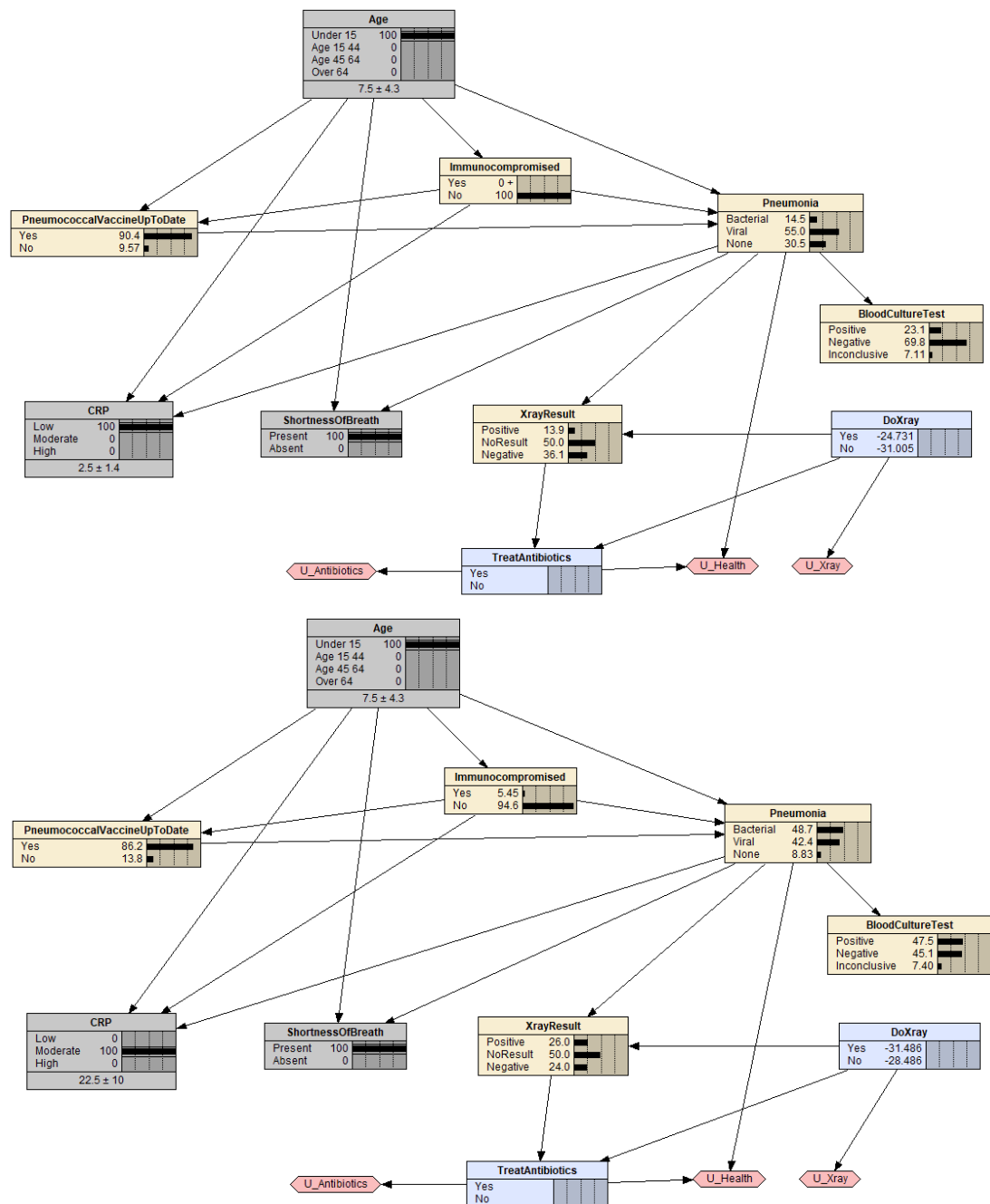


Given the patient did not do X-ray, the expected utility for taking antibiotics is -31.009 and for not taking the antibiotics is -24.317.



ii) The results are as follows:

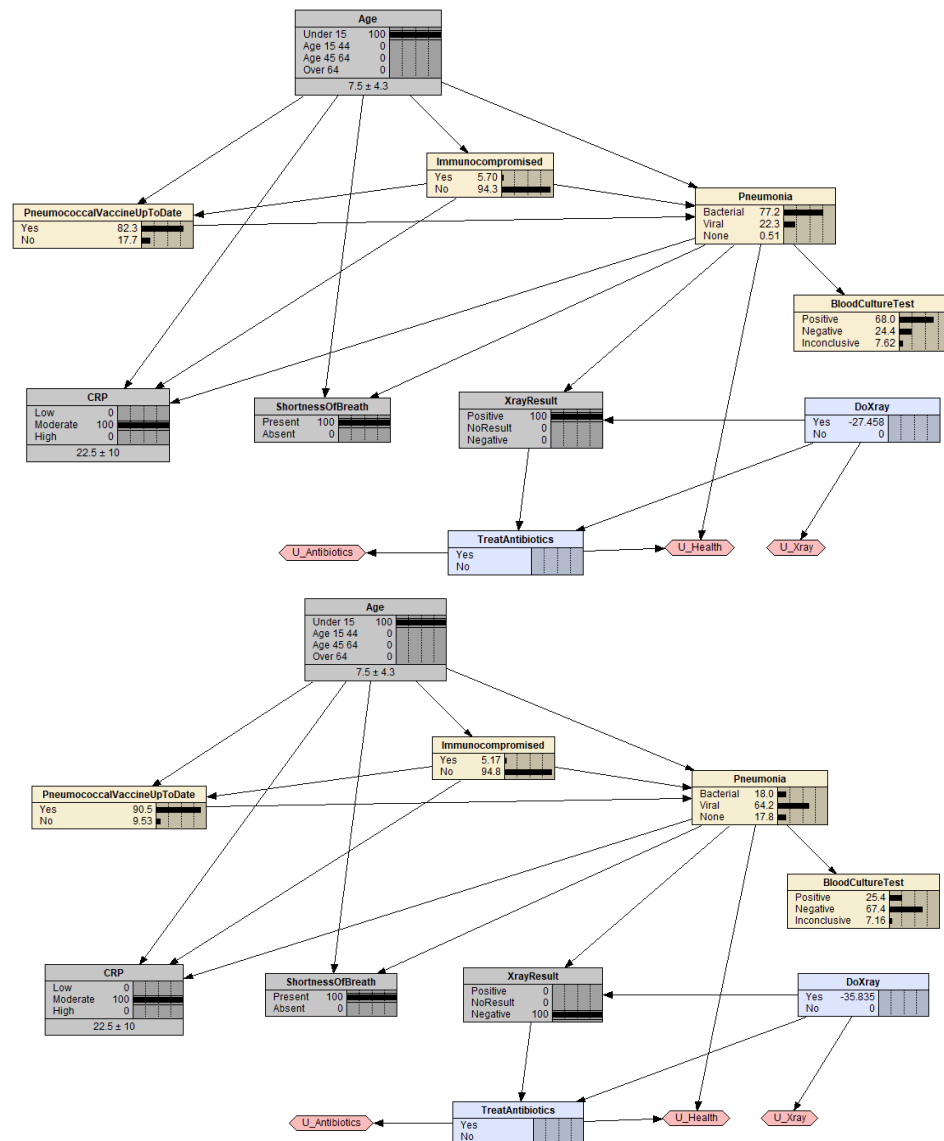
Evidence	P(Bacterial)	EU(X-ray)	EU(no X-ray)	Gap	Decision
CRP low	6.65%	-20.702	-24.317	+3.62	Take X-ray
CRP low + Shortness of breath	14.1%	-24.721	-31.005	+6.28	Take X-ray
CRP moderate + Shortness of breath	48.7%	-31.486	-28.486	-3.00	No X-ray



The utility I assigned in the previous question says the patient should only take treatment if $b > A/B = 10\%$. This means a patient should be prescribed when the chance of bacterial pneumonia is above 10%.

If we take the X-ray and the result is positive, chance will rise to 77.2%, we prescribe. If we take the X-ray and the result is negative, chance will fall to 18%, we still prescribe.

Since there is no result that would make the patient do anything different, doing an X-ray in this case would not be worth it.



Without the X-ray result, we should treat the patient with antibiotics. With shortness of breath and moderate CRP, the patient's probability of bacterial pneumonia is 48.7%, which is far above the 10% threshold. In addition, regardless of the X-ray result, we always have to prescribe as a negative result only drags the bacterial probability down to 18%, which is still above the 10% threshold.